

02477 – Bayesian Machine Learning: Lecture 7

Michael Riis Andersen

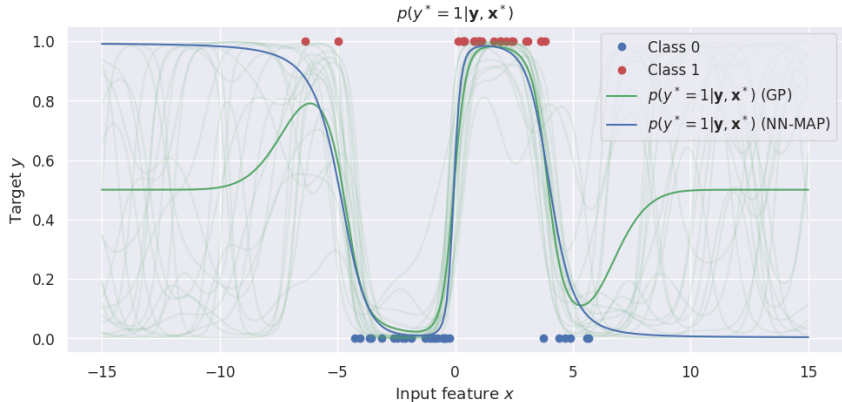
Technical University of Denmark,
DTU Compute, Department of Applied Math and Computer Science

Outline

- 1 Gaussian processes and neural networks
- 2 Generalized linear models and non-Gaussian likelihoods
- 3 Generalization and evaluation
- 4 Decision theory
- 5 Calibration

Gaussian processes and neural networks

From last week's exercise



- Some NN researchers are striving to make NNs behave more like Gaussian processes
- Some GP researchers are striving to make GPs flexible and as easy to scale as NNs
- Exploring relationships between GPs and NNs

Infinitely wide neural networks I

Exploring connections between GPs and NNs

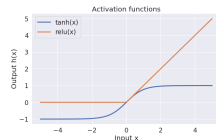
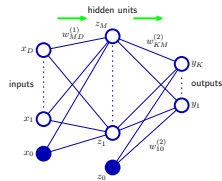
- Consider a network with a single hidden layer with H neurons and one output

$$z(x) = h(W_1 x + b_1)$$

$$f(x) = W_2 z(x) + b_2$$

- For $x \in \mathbb{R}^D$, then $W_1 \in \mathbb{R}^{H \times D}$ and $W_2 \in \mathbb{R}^{H \times 1}$

- *Universal approximators* for many classes of functions



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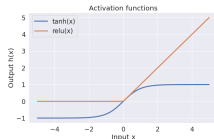
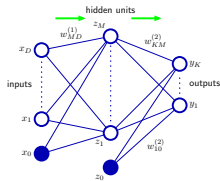
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- Let's now make some assumptions

1. Activation function h is bounded, e.g. $h(x) = \tanh(x)$
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4. Prior variance of $\sigma_w^2 = \frac{1}{H}$



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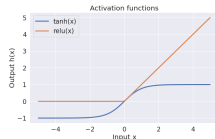
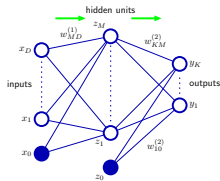
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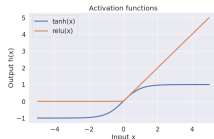
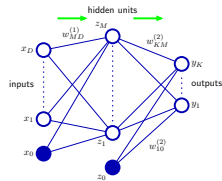
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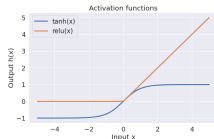
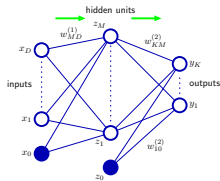
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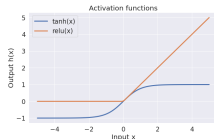
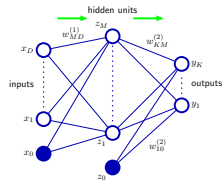
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- CLT implies the neural network $f(\mathbf{x})$ converges to a Gaussian process

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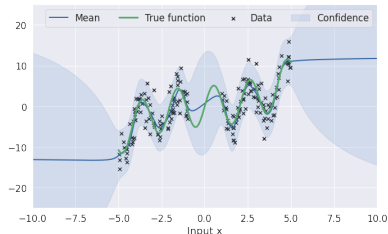
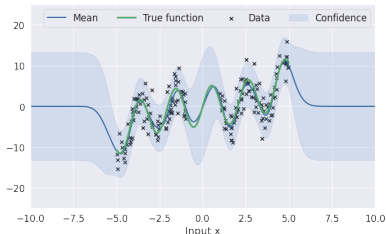
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- Comparing *squared exponential* and *neural network kernels*



Generalized linear models and non-Gaussian likelihoods

Likelihoods as observation models: Why bother?

- Consider a linear model

$$f(\mathbf{x}) = \phi(\mathbf{x})^T \mathbf{w}$$

- Regression using Gaussian likelihood

$$y_n | f_n \sim \mathcal{N}(y_n | f_n, \beta^{-1})$$

- Binary classification with sigmoid function $\sigma(\cdot)$

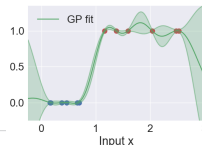
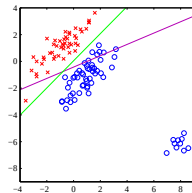
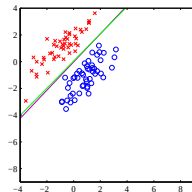
$$y_n | f_n \sim \text{Ber}(y_n | \sigma(f_n))$$

- ..., but why not just use regression for everything?

- More general setting: *generalized linear models* GLMs

$$y_n | f_n \sim p(y_n | f_n)$$

Least squared vs logistic regression



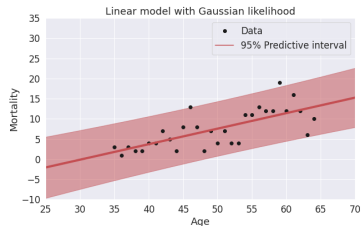
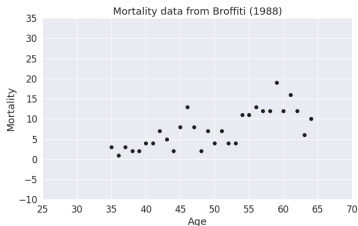
Common likelihoods in machine learning

Likelihood	Support	Example
Gaussian	$y_n \in \mathbb{R}$	Standard regression
Student's t	$y_n \in \mathbb{R}$	Standard regression with heavier tails
Exponential	$y_n \in (0, \infty)$	Strictly positive target values
Gamma	$y_n \in (0, \infty)$	Strictly positive target values
Bernoulli	$y_n \in \{0, 1\}$	Binary classification
Binomial	$y_n \in \{0, 1, \dots, N\}$	Sequence of Bernoulli trials
Poisson	$y_n \in \{0, 1, 2, 3, \dots, \}$	Count data
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Example: analyzing mortality as a function of age



Under this model, the posterior predictive distribution for age = 25 is $\mathcal{N}(-2.61, 14.76)$ meaning that $p(\text{mortality} < 0 | \text{age} = 25) \approx 0.75$

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Spend 5 minutes on the DTU Learn quiz: "Lecture 7a: Common likelihoods in machine learning."

Discuss what type of observation model might be appropriate for ...

1. Predicting whether a student will pass a course (based on the student's previous grades)
2. Predicting the time it takes to finish a written exam
3. Predicting number of errors in a multiple choice exam
4. Predicting whether the student will fill out the exam with a blue, red or black pen

Generalized linear models in three steps

The components of a generalized linear model (GLM)

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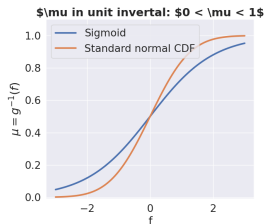
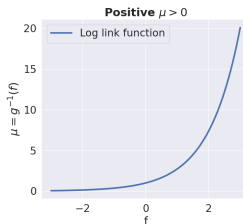
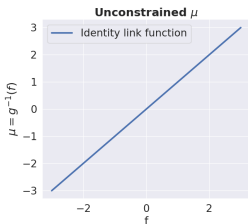
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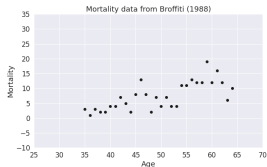
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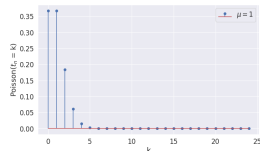
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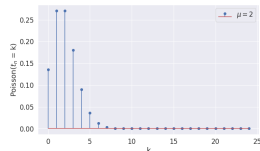
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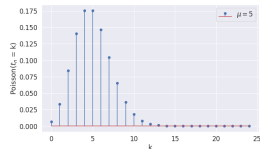
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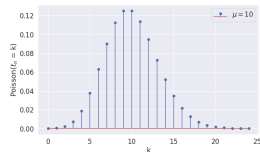
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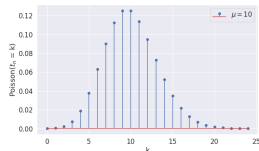
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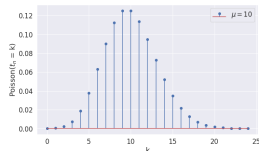
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- We use a Gaussian prior for $\mathbf{w}$

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \mathbf{I})$$



Example: Bayesian Poisson Regression II

- Using $\mu_n = \mu(\mathbf{x}) = \exp(\mathbf{x}_n^T \mathbf{w})$, the joint model becomes

$$p(\mathbf{y}, \mathbf{w}) = p(\mathbf{y}|\mathbf{w})p(\mathbf{w}) = \prod_{n=1}^N \text{Poisson}(y_n|\mu_n)\mathcal{N}(\mathbf{w}|\mathbf{0}, I)$$

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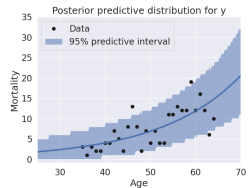
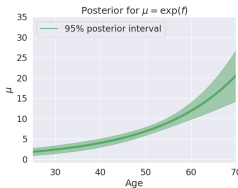
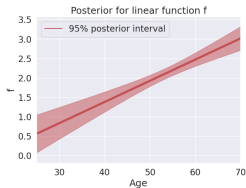
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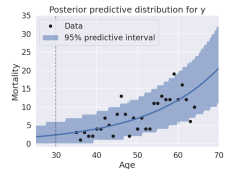
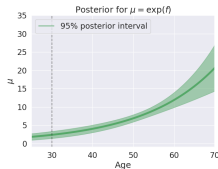
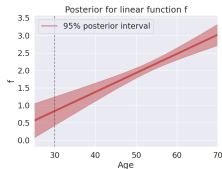
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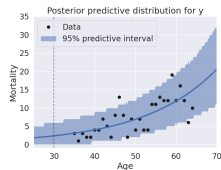
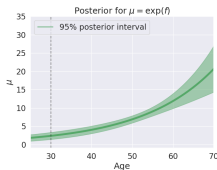
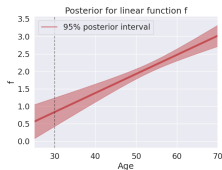
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- Calculating predictions for age = 30. Let $\mathbf{x}_* = [1 \quad 30]$, then $f_* = \mathbf{x}_*^T \mathbf{w}$

$$p(f_* | \mathbf{y}) = \mathcal{N}(f_* | \mathbf{x}_*^T \mathbf{m}, \mathbf{x}_*^T \mathbf{S} \mathbf{x}_*) \approx \mathcal{N}(f_* | 0.8, 0.2^2)$$



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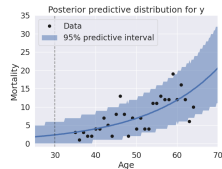
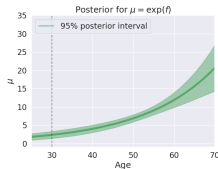
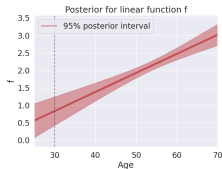
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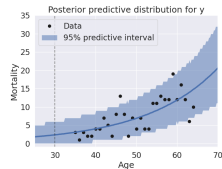
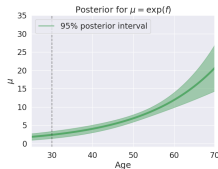
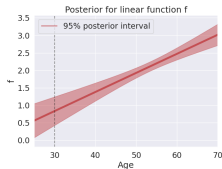
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1. Compute $\mu_*^{(s)} = \exp(f_*^{(s)})$
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- Finally, we calculate the sample means (or variances, percentiles etc)

$$\mathbb{E} [\mu_*^{(s)} | \mathbf{y}] \approx \frac{1}{S} \sum_{s=1}^S \mu_*^{(s)} = 2.34$$

$$\mathbb{E} [y_*^{(s)} | \mathbf{y}] \approx \frac{1}{S} \sum_{s=1}^S y_*^{(s)} = 2.34$$



Generalized GP/NN models in three steps

The components of a generalized model

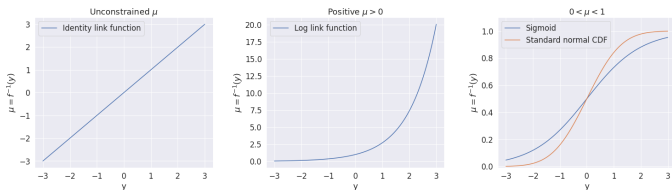
1. We replace the linear model with a Gaussian process (or a NN)

$$f(\mathbf{x}) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

2. The *link function* f that related the mean of the linear model to the mean of the response variable $t(\mathbf{x})$

$$g[\mu(\mathbf{x})] = f(\mathbf{x}) \quad \Longleftrightarrow \quad \mathbb{E}[y|\mathbf{x}] = \mu(\mathbf{x}) = g^{-1}[f(\mathbf{x})]$$

3. The distribution $p(y|\mathbf{x})$ for the response variable $y(\mathbf{x})$, e.g. Poisson, binomial, gamma etc.



Adapting GP models to different likelihoods

- From last weeks' exercise: Laplace approximation GP for classification

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- The gradient and Hessian of the log joint

$$\nabla_{\mathbf{f}} \log p(\mathbf{y}, \mathbf{f}) = \sum_{n=1}^N \nabla_{\mathbf{f}} \log p(y_n|f_n) - \mathbf{K}^{-1} \mathbf{f},$$

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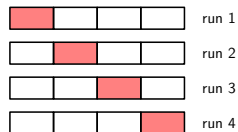
- Hence, all we need to change is the first and second order derivative of log likelihood

Generalization and evaluation

Generalization error and model evaluation

- Consider a supervised problem with $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$
- We measure performance of models using *cross-validation*
 1. Held-out test set
 2. K-fold
 3. Leave-one-out
 4. Split-half
 5. ...
- Goal: assess the *generalization error* of the model, i.e. how well can we expect the model to perform on a new, unseen test example?
- We use cross-validation to *estimate* the generalization error
- Parameter tuning: training/validation/test or nested cross-validation

4-fold cross-validation



Generalization and Capacity

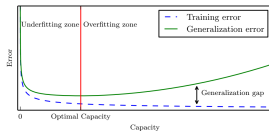


Figure from Goodfellow 2016

Generalization error and loss functions

- Let y be the target value for a input vector $\mathbf{x}$ and let $\hat{y} \equiv \hat{y}(\mathbf{x})$ be a prediction
- The *loss function* $\mathcal{L}(y, \hat{y})$ define the cost of predicting $\hat{y}$ when the true value is y

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- We rarely know the *true data generating mechanism* $p(\mathbf{x}, y)$ in practice

Estimating the generalization

- The *exact* generalization error

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- For ERM, one can show that the training error is *optimistic*

$$\mathbb{E} [\hat{\mathcal{R}}_{\hat{y}}^{\text{train}}] \leq \mathbb{E} [\hat{\mathcal{R}}_{\hat{y}}^{\text{test}}]$$

Example: the expected generalization error wrt. to the squared loss

- Consider the following (very simple) linear regression model (no slope, only intercept)

$$y = w_{\text{true}} + e, \quad e \sim \mathcal{N}(0, \sigma^2)$$

- We will use some estimator $\hat{w}$ for w_{true} (fixed) as a prediction for a new y , i.e. $y^* = \hat{w}$.
- What is the *generalization error* wrt. *squared loss*?

$$\mathcal{R}_{\hat{w}} = \mathbb{E}[\mathcal{L}] = \iint (t - \hat{w})^2 p(y, \mathbf{x}) dy d\mathbf{x}$$

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- Unsurprisingly, the generalization error is minimized when $\hat{w} = w_{\text{true}}$, which leads to $\mathbb{E}[\mathcal{L}] = \sigma^2$.

What is the expected generalization error for a given dataset size?

- The generalization error

$$\mathcal{R}_{\hat{w}} = (w - \hat{w})^2 + \sigma^2$$

- Suppose we observe a dataset $\mathcal{D} = \{y_n\}_{n=1}^N$ with the goal of estimating the parameter w_{true}

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- Interpretation

1. Average excess error due to finite training data
2. Inherent noise in the data

Decision theory

Uncertainty in multi-class classification

- Categorical distributions for multi-class classification with K classes

$$y_n | \mathbf{f}_n \sim \text{Categorical}[\text{softmax}(\mathbf{f}_n)]$$

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- Example

$p(y^* = \text{cat} \text{img})$	$= 0.36$	$p(y^* = \text{healthy} \text{x-ray})$	$= 0.36$
$p(y^* = \text{dog} \text{img})$	$= 0.34$	$p(y^* = \text{pre-phase cancer} \text{x-ray})$	$= 0.34$
$p(y^* = \text{bird} \text{img})$	$= 0.30$	$p(y^* = \text{severe cancer} \text{x-ray})$	$= 0.30$

Quantifying uncertainty for multi-class classification

- The *confidence* of the posterior predictive distribution is defined as

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$$p(y^* = \text{cat} | \text{img}) = 0.50$$

$$p(y^* = \text{dog} | \text{img}) = 0.50$$

$$p(y^* = \text{bird} | \text{img}) = 0.0$$

$$\mathcal{C} = 0.5$$

$$\mathcal{H} \approx 0.69$$

$$p(y^* = \text{healthy} | \text{x-ray}) = 0.50$$

$$p(y^* = \text{pre-phase cancer} | \text{x-ray}) = 0.25$$

$$p(y^* = \text{severe cancer} | \text{x-ray}) = 0.25$$

$$\mathcal{C} = 0.5$$

$$\mathcal{H} \approx 1.04$$

Reject option: I don't know?

- For applications in medicine etc. it may be better to say "I don't know" rather than provide a prediction we don't really trust

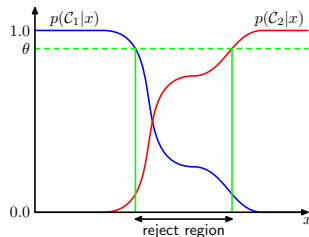
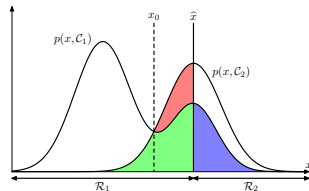
- **Reject option:** Avoid making a decision if the uncertainty is too large

- Reject to make decision if

$$\mathcal{C} = \max_k p(y^* = i | \mathbf{y}, \mathbf{x}^*) < \theta_{\text{reject}}$$

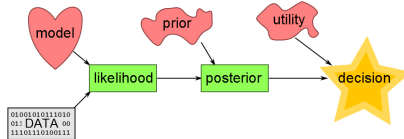
- If $\theta_{\text{reject}} = 1$, then all samples will get rejected

- If $\theta_{\text{reject}} \leq \frac{1}{K}$, then no samples will get rejected



Decision theory: a more formal approach

- In Bayesian modelling we strive to represent all unknown quantities (parameters, predictions etc.) using *probability distributions*
- Yet, we are often forced to reduce these to single decisions/predictions
 1. Cancer or not cancer?
 2. Which online campaign is better? A, B or C etc?
 3. How many airline passengers in year 2025?
 4. Which model best describes the data?
 5. ...
- *Statistical decision theory* tells us to make optimal decisions under uncertainty



Bayesian decision theory in a nutshell

- A decision maker has a set of actions/choices $a \in \mathcal{A}$ to choose from
- The optimal action depends on the true state $s \in \mathcal{S}$ of the system of interest
- The *loss function* $\mathcal{L}(s, a)$ determines the cost for choice a when the true state is s . We can also use a *utility function* $\mathcal{U}(s, a) = -\mathcal{L}(s, a)$

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- Example: assigning medical treatment to potential Covid19 patient

$\mathcal{L}(s, a)$	Do nothing	Isolate at home	Hospitalization
Healthy	0	10	50
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- In practice, we do not know the true state s . What to do?
- *Bayesian decision theory*
 1. Compute posterior of the state s given data $\mathbf{y}$, i.e. $p(s|\mathbf{y})$
 2. Choose the action that minimizes the posterior expected loss

$$\hat{a} = \arg \min_{a \in \mathcal{A}} \mathbb{E}_{p(s|\mathbf{y})} [\mathcal{L}(s, a)]$$

5 minutes exercise

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State s	Healthy	Mild symptoms	Severe symptoms
$p(s y)$	0.65	0.3	0.05

Questions

- What is the posterior expected loss for each action, i.e. $\mathbb{E}_{p(s|y)} [\mathcal{L}(s, a)]$?
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The best action is a_2 (to isolate at home) because it minimizes the posterior expected loss

Decision theory for classification I

- Consider a binary classification problem with $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$ for $y_i \in \{0, 1\}$
- The *0/1 utility function* is given by $\mathcal{U}(y, \hat{y}(\mathbf{x})) = \mathbb{I}[y = \hat{y}(\mathbf{x})]$

$\mathcal{U}(y, \hat{y}(\mathbf{x}))$	$\hat{y} = 0$	$\hat{y} = 1$
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- If we choose $\hat{y}(\mathbf{x}) = 0$, then

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Decision theory for classification I

- Consider a binary classification problem with $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$ for $y_i \in \{0, 1\}$
- The *0/1 utility function* is given by $\mathcal{U}(y, \hat{y}(\mathbf{x})) = \mathbb{I}[y = \hat{y}(\mathbf{x})]$

$\mathcal{U}(y, \hat{y}(\mathbf{x}))$	$\hat{y} = 0$	$\hat{y} = 1$
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- Picking the class with largest posterior pred. probability is *Bayes optimal* under the 0/1-loss function

Decision theory for classification II

- Consider a binary classification problem with $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$ for $y_i \in \{0, 1\}$
- Example: classifying cancer from medical imagery $\mathbf{x}$

$\mathcal{U}(y, \hat{y}(\mathbf{x}))$	$\hat{y} = 0(\text{not cancer})$	$\hat{y} = 1(\text{cancer})$
$y = 0(\text{not cancer})$	1	-10
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- We have

$$p \leq \frac{\mathcal{U}_{01} - \mathcal{U}_{00}}{\mathcal{U}_{10} - \mathcal{U}_{11} + \mathcal{U}_{01} - \mathcal{U}_{00}} = \frac{-10 - 1}{-100 - 1 - 10 - 1} \approx 0.099$$

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- That is, if $p(y^* = 0|\mathbf{x}) = 1 - p > 0.901$, we predict "No cancer"

Decision theory for regression I

- Consider a regression problem with $\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N$ for $y_i \in \mathbb{R}$

- The most common loss function for regression is the *quadratic loss*

$$\mathcal{L}(y, y(\mathbf{x})) = (y - \hat{y}(\mathbf{x}))^2$$

- The expected loss for a complete probabilistic description $p(\mathbf{x}, y)$ is

$$\mathbb{E}[\mathcal{L}(y, y(\mathbf{x}))] = \iint (y - \hat{y}(\mathbf{x}))^2 p(\mathbf{x}, y) d\mathbf{x} dy$$

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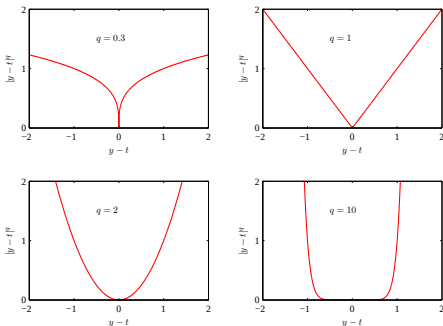
- What is the *optimal predictor function* $\hat{y}(\mathbf{x})$?

$$\hat{y}(\mathbf{x}) = \mathbb{E}[y|\mathbf{x}]$$

- The *posterior predictive mean* is *optimal* wrt. the quadratic loss

Decision theory for regression II

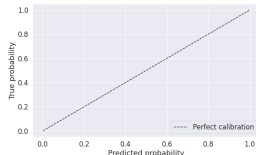
- The *Minkowski loss* is a generalization of the quadratic loss and is given by $\mathcal{L} = (\hat{y}(\mathbf{x}) - y)^q$
- One can show that ...
 - the posterior mean is optimal for $q = 2$
 - the posterior median is optimal for $q = 1$
 - the posterior mode is optimal for $q \rightarrow 0$
- $q = 2$ can be sensitive to outliers because of the quadratic form, while $q = 1$ is more robust



Calibration

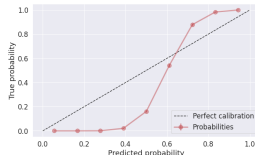
Calibration for classification models

- We have seen that probabilities play an important role in decision-making, but how do we know if the probabilities are accurate?
- Training models via maximum likelihood or Bayesian methods should in theory yield calibrated models, however, in practice, models are rarely perfectly calibrated
- If we have an independent validation/test set, we can quantify the degree of calibration or miscalibration
- Among all the example in the test set, where the predictive probability is approximately 80% we expect roughly 80% of the corresponding examples to belong to the positive class.
- Metrics for quantifying degree of miscalibration for classification
 1. Expected calibration error (ECE)
 2. Maximum calibration error
 3. Marginal calibration error
 4. Brier score



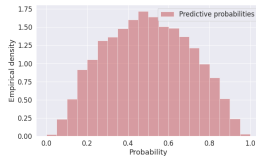
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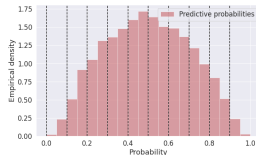
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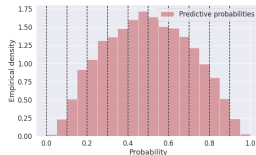
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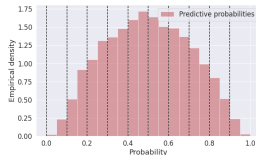
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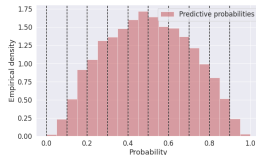


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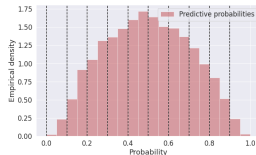
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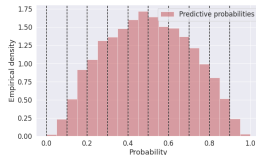
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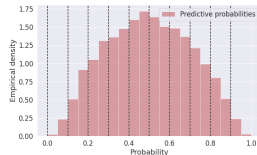
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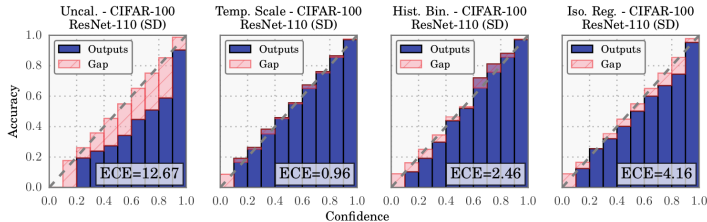
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- The *expected calibration error (ECE)* is then given by

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{M} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|$$



Expected calibration error (ECE)



- Results on CIFAR-100 dataset from Guo et al, 2017 (and page 573 in Murphy2)
- Several methods for recalibration:
 1. Platt scaling
 2. Temperature scaling
 3. Histogram binning
 4. Isotonic regression
- In *temperature scaling*, we introduce a temperature parameter $T > 0$ in the softmax-function:

$$p(y|f(\mathbf{x}), T) = \text{Cat}(y|\text{softmax}(f(\mathbf{x})/T))$$
- T is estimated using a validation set.

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- Bayesian decision theory: Uncertainties are crucial for optimal decision-making and therefore *calibration* becomes important
- The calibration error can be assessed via the *expected calibration error* (ECE) metric or *reliability curves*

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{M} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|$$